

AI-Powered Requirement Analysis & Quality Report (Assignment 2)

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WITHOUT CLARIFICATION (Baseline)	WITH AI CLARIFICATION (Refined)	MEASURED QUALITY DELTA
8 / 100 Poor (0% Testability)	96 / 100 Excellent (100% Testability)	+88 Points ISO 29148 Score Gain

1. Stakeholder Clarification Q&A; Audit Log

# / Category	Targeted Question & Stakeholder Decision
Q1 [Performance]	Q: What specific latency threshold defines acceptable performance? Stakeholder Answer: Single resume parsing must be under 1.5s (95th percentile) and batch upload of 100 resumes must complete within 30 seconds.
Q2 [Fairness]	Q: What quantitative fairness metric and audit frequency should be enforced? Stakeholder Answer: Disparate impact ratio between 0.80 and 1.25 across gender and college tiers; quarterly fairness audit.
Q3 [Accuracy]	Q: What objective accuracy metric defines "good enough for HR trust"? Stakeholder Answer: Top-10 candidate precision $\geq 85\%$ and NDCG@10 ≥ 0.82 evaluated against consensus of 3 senior recruiters.
Q4 [Explainability]	Q: How should candidate match reasoning be presented to recruiters? Stakeholder Answer: Interactive match scorecard showing matched skills %, project complexity score, and top 3 justification reasons.
Q5 [Ranking Algorithm]	Q: How should skills, experience, and projects be weighted? Stakeholder Answer: Formula: 45% skills match + 35% project complexity + 20% experience with tier normalization.
Q6 [Data Ingestion]	Q: What file formats and size limits must be supported? Stakeholder Answer: Support PDF and DOCX up to 10MB per file with automated text extraction into validated JSON schema.
Q7 [Project Scope]	Q: What is the committed delivery milestone for the MVP? Stakeholder Answer: 4-week MVP deliverable covering ingestion, JD matching, top-10 ranking, and scorecard export.

2. Refined Functional Requirements (FR)

ID / Title	Specification & Acceptance Criteria	Status / Source
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FR-01 Resume Ingestion & Parsing Engine	The system shall ingest resumes in PDF and DOCX formats (up to 10MB per file) and parse text into structured JSON schemas as specified by stakeholder: Support PDF and DOCX up to 10MB per file with automated text extraction into validated JSON schema.. Acceptance Criteria: - Given a valid PDF or DOCX resume under 10MB, When uploaded, Then the system extracts text and generates valid structured JSON. - Given a corrupted or unsupported file type, When uploaded, Then the system returns an informative HTTP 422 error.	RESOLVED STAKEHOLDER_CLARIFICATION
FR-02 Deterministic Multi-Factor Candidate Scoring Engine	The system shall calculate composite candidate relevance scores using stakeholder-defined weighting: Formula: 45% skills match + 35% project complexity + 20% experience with tier normalization.. Acceptance Criteria: - Given a job description and candidate profile, When scored, Then the final score strictly follows the stakeholder weighting formula. - Given an exceptional fresher with high project complexity, When ranked against generic experience, Then the fresher outranks when composite score is higher.	RESOLVED STAKEHOLDER_CLARIFICATION
FR-03 Structured Explainability & Match Scorecard	The system shall generate an interactive candidate match scorecard satisfying stakeholder preference: Interactive match scorecard showing matched skills %, project complexity score, and top 3 justification reasons.. Acceptance Criteria: - Given a shortlisted candidate, When viewed in the UI, Then the system displays matched skills %, project complexity score, and top 3 justification reasons. - Given recruiter requests match explanation, When requested, Then the explanation payload renders in < 500ms.	RESOLVED STAKEHOLDER_CLARIFICATION
FR-04 Role-Based Recruiter & Hiring Manager Dashboard	The system shall provide secure role-based portals for HR Recruiters and Hiring Managers to upload job descriptions, trigger bulk resume scoring, and export shortlists. Acceptance Criteria: - Given authenticated recruiter, When uploading JD, Then analysis job is dispatched. - Given unauthorized user, When accessing dashboard, Then HTTP 403 Forbidden is returned.	RESOLVED RAW_DIALOGUE

3. Refined Non-Functional Requirements (NFR)

ID / Category	Requirement Statement & Verifiable Metric	Verification Method
NFR-PERF-01 [Performance]	The system shall meet latency SLO: Single resume parsing must be under 1.5s (95th percentile) and batch upload of 100 resumes must complete within 30 seconds.. Target Metric / SLO: Single resume parsing must be under 1.5s (95th percentile) and batch upload of 100 resumes must complete within 30 seconds.	Automated Load Benchmark (k6/Locust) & Datadog APM Tracing
NFR-FAIR-01 [Fairness]	The ranking algorithm shall enforce demographic fairness standard: Disparate impact ratio between 0.80 and 1.25 across gender and college tiers; quarterly fairness audit.. Target Metric / SLO: Disparate impact ratio between 0.80 and 1.25 across gender and college tiers; quarterly fairness audit.	Automated Fairness Audit Suite (AIF360/Fairlearn) & PII Redaction Test

NFR-ACC-01 [Accuracy]	The candidate shortlisting model shall achieve accuracy benchmark: Top-10 candidate precision >= 85% and NDCG@10 >= 0.82 evaluated against consensus of 3 senior recruiters.. Target Metric / SLO: Top-10 candidate precision >= 85% and NDCG@10 >= 0.82 evaluated against consensus of 3 senior recruiters.	Offline Evaluation Benchmark & Human Recruiter Ground-Truth Test Set
NFR-SCOPE-01 [Scope]	The project engineering team shall deliver the validated MVP deliverable compliant with: 4-week MVP deliverable covering ingestion, JD matching, top-10 ranking, and scorecard export.. Target Metric / SLO: 4-week MVP deliverable covering ingestion, JD matching, top-10 ranking, and scorecard export.	Sprint Milestone Verification & Release Checklist Sign-off
NFR-SEC-01 [Security]	Candidate personally identifiable information (PII) must be stored with AES-256 encryption at rest and TLS 1.3 in transit. Target Metric / SLO: AES-256 encryption at rest, TLS 1.3 in transit, GDPR Article 32 compliant	Automated Security & Encryption Audit
NFR-USAB-01 [Usability]	The recruiter dashboard interface shall achieve System Usability Scale (SUS) score >= 80.0. Target Metric / SLO: System Usability Scale (SUS) >= 80.0	User Usability Testing

4. Quality Comparison Audit Matrix (ISO 29148 Principles)

Quality Dimension	Without Clarification	With Clarification	Improvement (Delta)
Overall Quality Index (OQI)	8/100	96/100	+88 pts
Ambiguity Score	100%	0%	-100% reduction
Testability & Verifiability	0%	100%	+100% gain
Completeness Coverage	0%	100%	+100% gain
Specificity & Measurability	0%	90%	+90% gain
Traceability & Lineage	50%	85%	+35% gain